Calculator [22] for this purpose, integrating parameters like GPU power consumption (derived from the NVIDIA System Management Interface), model training time, chosen cloud provider, and the geographic locale of computation. For our evaluations, we selected the Google Cloud Platform as the cloud provider and the U.S. as the computational region (with an estimated carbon footprint of 566.3 gCO2/kWh), as shown in Section IV.

IV. Results

In this section, we analyze the data generated in Section III to answer our questions of interest. As can be seen from Figure 1a, FLOP counts of our training runs span two orders of magnitude, and the generated data is sufficiently diverse.

A. What Algorithmic Choices are Driving Adversarial Robustness Advances?

In this section, we aim to unravel which algorithmic interventions are important for advancing adversarial robustness. We first try to predict how well a model (using a specific adversarial training recipe) can perform without actually training or testing it. Specifically, we use number of model parameters, synthetic data $\in$ {True, False}, activation $\in$ {ReLU, GELU}, loss $\in$ {PGD, TRADES}, PGD steps $\in$ {1, 2, 5, 7, 10}, and exponential moving average (EMA) $\in$ {True, False} as input features to our predictive model. We use gradient boosting regression model with number of estimators equal to 50 and max depth equal to 5. We use a 70:30 train-test split in our experiment.

In Figure 1b, we plot AutoAttack accuracy for unseen/holdout adversarial training configurations predicted by our model against ground truth values. We see that our model can predict the performance of unseen test configurations with high accuracy.

To the best of our knowledge, this is the first attempt to show that the adversarial robustness of a given training recipe can directly be predicted without going through a computationally expensive training-validation cycle.

This finding is expected to have broad implications for compute-efficient training [4] and reliable testing [47] of machine learning models.

Next, knowing which algorithmic knobs are important for predicting AutoAttack accuracy can be very useful in helping designers understand the problem and devising better approaches for improving the adversarial robustness. Thus, we analyze the feature importance of our predictive model in Figure 1c. Feature importance provides a score that indicates how useful (or valuable) each feature is for our predictive model. These importance scores are calculated individually for each feature in our dataset, enabling the features to be ranked and compared to each other in Figure 1c.

We hypothesize that the use of synthetic data, training loss, and the number of model parameters are the most important driving forces behind the adversarial robustness advances.

We take a closer look into this finding in Figure 2. We see that in the data-limited regime, i.e., without synthetic data, the choice of the loss function (PGD vs. TRADES) or regularization (EMA vs No EMA) plays an important role in improving the adversarial robustness of a given model – evidenced by high variance in autoattack accuracy. However, in the presence of synthetic data, the variance due to these choices is reduced, and model size plays a greater role. These findings are reasonable and also evident from popular leaderboards such as RobustBench [7].

B. What is the Computational Burden of Adversarial Robustness?

Having identified promising strategies for improving adversarial robustness in Section IV-A, in this section, we explore the limits of scaling compute (i.e., FLOPs) to achieve adversarial robustness. Specifically, we study empirical scaling laws for adversarial robustness performance in terms of AutoAttack accuracy. We first divide the FLOP range into 19 equisized bins. Then, for each FLOP bin, we determine which run achieves the best AutoAttack accuracy. This serves as the best performance envelope for our FLOP range in consideration. Similarly to Thompson et al. [38], we expect a polynomial relationship between compute and AutoAttack accuracy:

$$\text{accuracy} = C \times (\text{FLOPs})^{\alpha}, \tag{4}$$

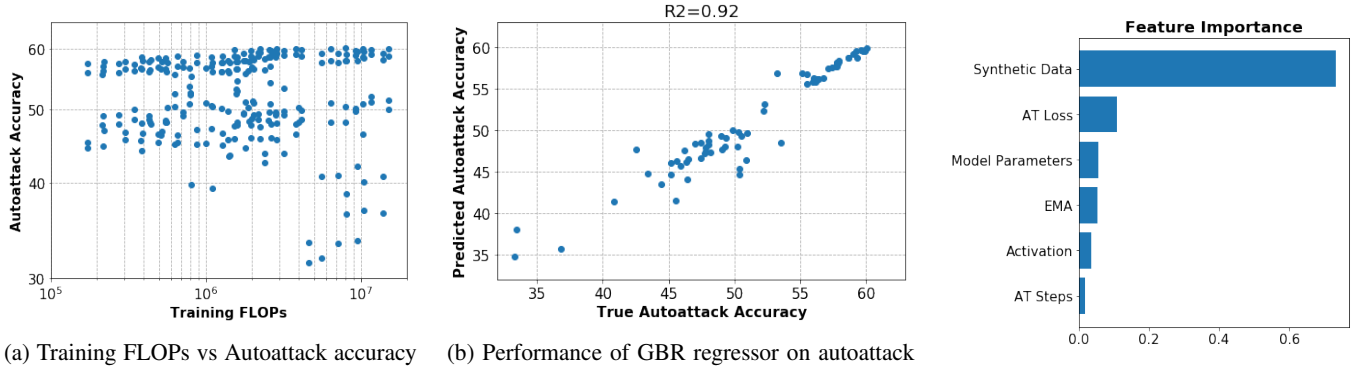
where C and α are learnable parameters. Next, we fit power laws to estimate the best AutoAttack accuracy for any given amount of compute.

Figure 3a shows the growth in the AutoAttack accuracy as a function of the computation used in these models. Given that this is plotted on a log-log scale, a straight line indicates a polynomial growth in computing per unit of performance. In other words, there is a clear power law relationship between the adversarial robustness and compute spanning two orders of magnitude (with $R2 = 0.76$). We find that the power law relationship also holds for electricity cost (see Figure 3b) and CO2 emission (see Figure 3c) both with $R2 = 0.95$.

AutoAttack accuracy, electricity cost, and CO2 emission of adversarial training follow a power law with respect to compute (i.e., FLOPs).

Unfortunately, the growth rate for AutoAttack accuracy is very small as seen from the estimated parameters $\alpha^* = 0.01$ and $C^* = 50.86$. On the other hand, the growth rate for CO2 emission (electricity cost) is quite large, i.e., $\alpha^* = 0.95$ with $C^* = 7.0 \cdot 10^{-6}$. These findings suggest that scaling compute alone may not be able to help solve the adversarial robustness problem.

Slow growth rate (w.r.t. compute) for adversarial robustness but high growth rates for CO2 emission and electricity cost imply that scaling compute is neither effective nor efficient.



(a) Training FLOPs vs Autoattack accuracy in log-log space.

(b) Performance of GBR regressor on autoattack accuracy prediction task.

(c) Feature importance of GBR model.

Fig. 1: Analysis of various algorithmic changes on adversarial robustness.

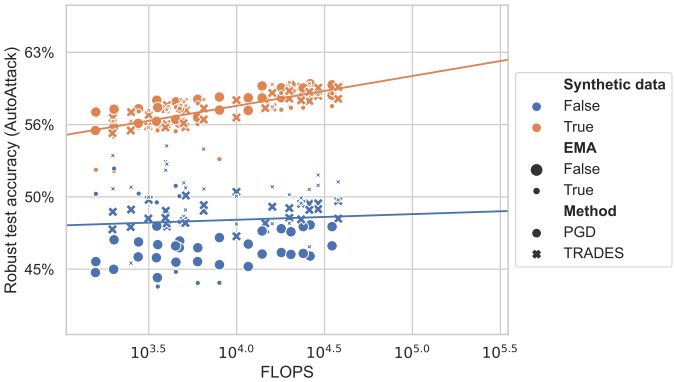


Fig. 2: A deeper look into the effect of various algorithmic changes on adversarial robustness.

Conversely, it has been observed that, for standard training, much higher growth rates hold hence suggesting that scaling compute is much more effective than it is for adversarial training.

Finally, we note, from Figure 2, that *the usage of extra data (despite being synthetic) has significantly more advantageous scaling laws*, compared to the models trained without extra data.

C. Are Adversarial Training Techniques Brittle?

Careful readers may have noted, at this point, that the best model among those in Figure 1a has only little more than 60% accuracy (to be precise, 60.14%). They may also have noted that, among the setups we test, there is also the training setup used by Gowal et al. [15], i.e., WideResNet-70-16 with GELU activation function, 1M synthetic samples, TRADES loss with 10 steps attack, 400 epochs, and EMA. This model has a reported 63.58% AutoAttack accuracy on the original paper, while our model, trained with the same set-up, has 59.92% robust accuracy, more than 3.6% less than the results reported in the paper.

We implemented the adversarial training pipeline from scratch, starting from the `timm` library[3] and by adding components based on the descriptions in the papers we reproduce, as well as by observing the code released with the papers. We also followed the suggestions made by Rade [30] (e.g., not updating mean and variance of the BatchNorm layers statistics when computing the adversarial example) to match as much as possible the original set-up by [15]. We double-checked our implementation and setups, including code, architectures, and hyper-parameters carefully, without any luck in exactly matching the results from Gowal et al. [15].

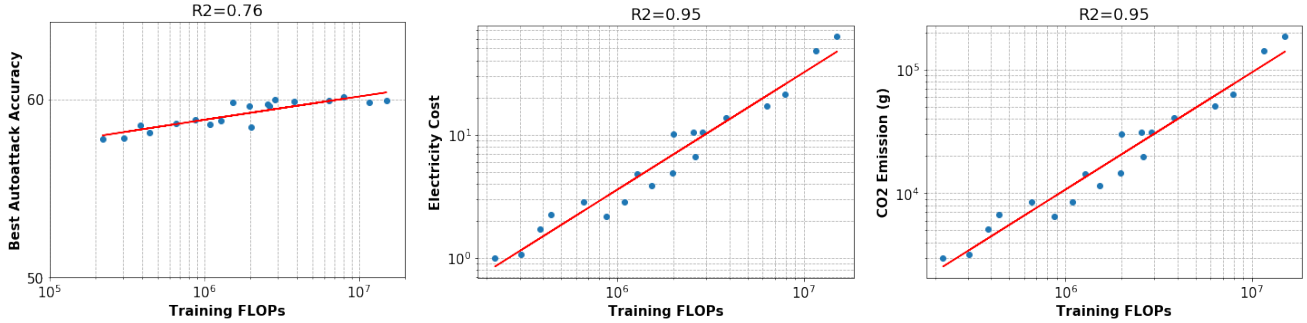
While we admit that exactly reproducing those results may not be impossible, it turned out to be very challenging. The fact that it is not possible to reproduce these results suggests that these methods may be relying on relatively brittle features, that are not really robust to small changes in the training setup. Hence, we believe that one factor to keep in account when trying to improve adversarial robustness before scaling up compute is to make sure to rely on setups that are robust to small changes in the training hyperparameters.

V. Discussion and Limitations

Our work presents several limitations, including the fact that we do not match exactly the results from previous work, we are only testing on a low-resolution and small-sized dataset, and we do not have confidence intervals. However, none of these limitations is easy to address in practice: we spent non-negligible time (and compute resources) to reproduce the results from the literature, and scaling up to larger datasets such as ImageNet and doing multiple runs per sample would be impractical in terms of compute (training WideResNet-76-10 for 400 epochs with 10-steps TRADES takes more than 16 days on a node with four Nvidia V100 GPUs).

Regardless, we believe that our work still gives a clear overview that scaling up compute does not give as much advantage for adversarial training as it does for standard training, especially without the usage of extra data. Thus, future research should explore the generation and the usage of extra data for improving adversarial training. We advocate for

[3]https://github.com/huggingface/pytorch-image-models



(a) Best autoattack accuracy follows a power law with a small growth rate.

(b) Electricity cost follows a power law with a high growth rate.

(c) CO2 emission follows a power law with a high growth rate.

Fig. 3: Limits of scaling compute on adversarial robustness by analyzing best autoattack accuracy configurations in log-log space.

more innovative and comprehensive approaches for continued progress in improving adversarial robustness, encouraging the field to explore new directions. We firmly believe that devising new methods to achieve robust and resilient AI systems is not only necessary but also holds promise for a future where machine learning technologies can withstand adversarial threats.

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